

# Chapter 42

## The remote and the impossible in Serbian

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Conditionals characterize epistemic possibilities (O-marked conditionals) or states of affairs that are somehow remote or impossible (X-marked conditionals). Serbian uses a combination of different verbal moods, conditional connectives, and tense marking to encode the relevant distinctions. We provide a compositional analysis that treats interpretation as sensitive to a salient belief state (by default, a representation of what is considered epistemically possible in the ongoing conversation). The salient belief state determines mood marking (indicative vs. conditional), and can be shifted by the connective *da*. Additionally, perfect in *da*-marked antecedents can introduce a temporal backshift to "rerun history". We argue that Serbian provides empirical evidence for a split between modal and temporal X-marking as proposed in the recent formal semantic literature, and discuss extensions of our assumptions to other occurrences of the relevant markers (complement clauses, purposive clauses, and standalone uses of dependent marked clauses).

**Keywords:** conditionals, mood, X-marking, subjunctive

### 1 Conditionals and X-marking

Conditional clauses serve to further characterize states of affairs that are either epistemic possibilities in the ongoing conversation, or are in some sense and to some degree remote from what counts as epistemically possible. Conditionals of the first kind typically appear with indicative morphology on both antecedent and consequent, whereas conditionals about somehow remote states of affairs usually include specialized verbal moods, or seemingly repurposed ('fake', [Iatridou 2000](#)) tense or aspect morphology. The standard terminology often relies

on morphosyntactic categories ('subjunctive', 'simple past',...) that can hinder crosslinguistic comparison, or semantic categories ('counterfactual', 'potentialis', 'realis', 'irrealis',...) that pre-empt investigations of what is conventionally encoded. To avoid either pitfall, von Fintel & Iatridou (2023) propose novel terminology, distinguishing between O(RDINARY)-MARKED and (E)X(TRA)-MARKED CONDITIONALS. For English, we obtain a contrast between O-marked indicative conditionals as in (1) and at last two types of X-marked conditionals relying on repurposed past and past perfect morphology as in (2).

- (1) If they invite me to that party tomorrow, I will go.  
O(RDINARY)-MARKED, indicative conditional
- (2) a. If they invited me to that party tomorrow, I'd go.  
SIMPLE PAST (SP), X-MARKED; present-counterfactuals  
b. If they had invited me to that party tomorrow, I'd have gone.  
PAST PERFECT (PP), X-MARKED; past-counterfactuals

The distinction between O-marking and X-marking has prompted long-standing theoretical debates concerning the particular nature of 'remoteness' indicated by the latter and whether it is the same across different X-marking constructions. While the default assumption is probably that the same one or more remoteness concepts are relevant across languages, it is unclear how different languages encode these concepts compositionally. In this respect it is of particular interest whether X-marking like English 'fake' past should be semantically unified with regular (in this case, temporal) uses of the same marker.

The purpose of this paper is to investigate X-marking in Serbian, a language that, as we argue, has the potential to shed novel light on this phenomenon. Unlike the X-marking systems in other languages that have been studied in detail so far, Serbian combines (i) choice of connective, (ii) verbal mood, (iii) temporal morphology to express different types of remoteness. In this, Serbian transparently splits modally and temporally interpreted X-marking along the lines of what has been hypothesized in the recent literature.<sup>1</sup>

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<sup>1</sup>Across languages, the X-markers found in conditional clauses tend to also appear on modals and desire predicates (Iatridou 2000; von Fintel & Iatridou 2023; Mizuno 2023; Sæbø 2024, a.o.). While the semantic effects found in these other contexts are intuitively related to the ones found in conditionals, the details are still under debate. Interestingly, the relevant literature contains little explicit discussion of different types of X-markers (as exemplified by English SP and PP-conditionals) outside of conditionals (but see Ferreira 2023; Mizuno 2023). While X-marking appears also on Serbian priority modals and desire predicates, we focus on conditional clauses in this paper, leaving to future research an extension to these other domains.

## 2 Basics of Serbian X-marking

### 2.1 Background: Conditionals

Semantic theories largely agree that conditionals instantiating the schema ‘If  $p$  (= antecedent),  $q$  (= consequent)’ express that  $q$  is true at all evaluation points in the relevant domain that make true  $p$ , and mark or constrain what domain supplies the relevant  $p$ -indices. Recent theories distinguish three domains that roughly correspond to three degrees of remoteness in the descriptive literature (e.g., realis–potentialis–irrealis) (Schulz 2017; Mizuno & Kaufmann 2019; von Prince 2019, a.o.):

*D1: the speaker’s / participants’ epistemic alternatives at the time of evaluation*

D1 is the quantificational domain of O-marked (‘indicative’) conditionals like (1).

*D2: epistemic alternatives (D1) together with relevant remote or impossible worlds*

D2 is the quantificational domain of English simple-past X-marked conditionals like (2a).

*D3: worlds that can be ruled out based on current knowledge, but were live possibilities at an earlier point (‘how things could have turned out’)*

D3 is the quantificational domain of English past perfect X-marked conditionals like (2b).

Note that, as reflected in the traditional terminology of ‘potentialis’ vs. ‘irrealis’, the form types that we associate with D2 and D3, respectively, are often distinguished by *degrees* of remoteness (‘far-fetched, but possible’ vs. ‘impossible’). We follow recent works that characterize the distinction in terms of *how* remote possibilities are reached; we will explain and motivate this choice in Section 3.3.

### 2.2 The Serbian X-marking Pattern

The examples in (3)–(5) showcase that Serbian O- vs. X-marking employs a combination of three ingredients: (i) an analytic mood “conditional” (aorist of *biti* ‘be’ + past participle of lexical verb); (ii) a switch in complementizer (*ako/da*); and (iii) present vs. perfect in *da*-antecedents (D2<sub>da</sub> vs. D3). D2<sub>ako</sub> and D2<sub>da</sub> are synonymous.<sup>2</sup>

<sup>2</sup>The interpretation of the different types of D2 conditionals appears to be subject to speaker variation. An anonymous reviewer describes D2<sub>da</sub> as more remote than D2<sub>ako</sub>. We have not

(3) D1-CONDITIONALS:

- a. Ako me **pozovu** na žurku, **ići ću**.  
 AKO I.DAT invite.3PL.PRS on party go.INF will.1SG  
 ‘If they invite me to the party, I’ll go.’

(4) D2-CONDITIONALS:

- a. Ako **bi** me **pozvali** na žurku, **išla bih**.  
 AKO be.3PL.AOR I.DAT invited.M.PL on party gone.F.SG be.1SG.AOR  
(D2<sub>ako</sub>)
- b. Da me **pozovu** na žurku, **išla bih**.  
 DA I.DAT invite.3PL.PRS on party gone.F.SG be.1SG.AOR  
(D2<sub>da</sub>)  
 ‘If they invited me to the party, I’d go.’

(5) D3-CONDITIONALS

- a. Da **su** me **pozvali** na žurku, **išla bih**.  
 DA be.3PL.PRS I.DAT invited.M.PL on party gone.F.SG be.1SG.AOR  
 ‘If they had invited me to the party, I’d have gone.’

Table 1 summarizes the strategies for distinguishing D1, D2, and D3-style conditionals in Serbian.<sup>3</sup>

The distinction between perfective and imperfective aspect does not seem to play a role for X-marking in Serbian. The aspectual oppositions follow their regular meaning contributions in conditional antecedents, as exemplified for D3-conditionals in (6)-(7).

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been able to confirm this judgment. To tap into possible effects along these lines, we also considered conditionals that progress from more normal to less normal kinds of antecedent situations and ascribe them different properties (SOBEL SEQUENCES, Lewis 1973). The remoteness differences between D1 and D2 in English is reflected in a felicitous progression from D1 to a more remote situation presented in D2 (i.a), while the opposite in (i.b) is infelicitous:

- (i) a. If we jump out of the window, we die. But if we jumped and someone with a big mattress happened to walk by just then, then we might be ok.  
 b. #If we jumped out of the window, we would die. But if we jump and someone with a big mattress happens to walk by just then, then we might be ok.

The reviewer’s intuition would lead us to expect a preference for a progression from D2<sub>ako</sub> to D2<sub>da</sub>, with the reverse being infelicitous. Neither the second author of this paper nor the speakers we consulted report an effect along these lines.

<sup>3</sup>Serbian counterfactual mood is related to the Russian marker *by*, but the overall X/O-marking patterns of the two languages differ in interesting ways that we cannot explore in this paper; see e.g. Jakob (2005); Sæbø (2024) for aspects of X-marking in Russian.

Table 1: The Serbian X-marking paradigm

| Type | Connective | Antecedent                          | Consequent                          |
|------|------------|-------------------------------------|-------------------------------------|
| D1   | <i>ako</i> | indicative (present)                | indicative (future)                 |
| D2   | <i>ako</i> | conditional ( <i>bi</i> +past.part) | conditional ( <i>bi</i> +past.part) |
|      | <i>da</i>  | indicative (present)                | conditional ( <i>bi</i> +past.part) |
| D3   | <i>da</i>  | indicative (perfect)                | conditional ( <i>bi</i> +past.part) |

- (6) Da je Jovan umro, sahranili bismo ga na  
 DA be.3SG.PRS Jovan died.PFV.M.SG, buried.M.PL be.1PL.AOR him on  
 planini.  
 mountains  
 ‘If Jovan had died, we’d have buried him in the mountains.’

- (7) Da je Jovan umirao, sa njim bi bio  
 DA be.3SG.PRS Jovan died.IPFV.M.SG with him be.3SG.AOR been.M.SG  
 doktor.  
 doctor.  
 ‘If Johan had been dying, the doctor would have been with him.’

In this respect, Serbian patterns with Russian and Polish, unlike e.g., Greek and Hindi, which employ ‘fake imperfective’ (see [Iatridou 2000; 2009](#), a.o.).

### 2.3 Further types of conditional constructions

While we take the pattern in Table 1 to be representative and possibly exhaustive for the remoteness distinctions reflected as O- vs. X-marking, we want to emphasize that these form types do not exhaust the possibilities for expressing conditionals in Serbian in general. For instance, *ako* can be replaced with other connectives like *ukoliko* or *kada*:<sup>4</sup>

- (8) a. {Ukoliko / kada} me pozovu na žurku, ići ću.  
 in.case / when I.DAT invite.3PL.PRS on party go.INF. will.1SG  
 roughly: ‘If they invite me to the party, I’ll go.’ (D1)

<sup>4</sup>The difference between *ukoliko* ‘in case’ and *ako* in D1 and D2 conditionals is reminiscent of the one between English *if* and *in case*, or German *wenn* and *falls* (see [Kaufmann et al. 2024](#)). *kada* is also an interrogative pronoun (‘when’), and may allow an additional construal as a temporal adverbial clause in the indicative marked (8a) (i.e., ‘at the time when...’). While extremely interesting, these contrasts are beyond the scope of our investigation.

- b. {Ukoliko / kada} bi me pozvali na žurku, išla  
 in.case / when be.3PL.AOR I.DAT invited.M.PL on party gone.F.SG  
 bih.  
 be.1SG.AOR  
 roughly: ‘If they invited me to the party, I’d go.’ (D2)

Antecedents can be marked with fronting of the verb + interrogative marker *li*:

- (9) Pozovu li me na žurku, {ići ću / išla bih}.  
 invite.3PL.PRS Q I.DAT on party go.IMP.2SG / gone.F.SG be.1SG.AOR.  
 ‘In case / when they invite me to the party, I’ll go / I’d go.’ (D1/D2)

As in other languages investigated for this property (and see [Arsenijević 2021](#) for Serbian specifically), consequents can be imperative clauses (only D1 and D2), see (10), or interrogative clauses, see (11):

- (10) a. Ako te pozovu na žurku, idi!  
 AKO you.DAT.SG invite.3PL.PRS on party go.IMP.2SG  
 ‘If they invite you to the party, go!’ (D1)
- b. Ako bi te pozvali na žurku, idi!  
 AKO be.3PL.AOR you.DAT.SG invited.M.PL on party go.IMP.2SG  
 ‘If they invited you to the party, go!’ (D2<sub>ako</sub>)
- c. Da te pozovu na žurku, idi!  
 DA you.DAT.SG invite.3PL.PRS on party go.IMP.2SG  
 ‘If they invited you to the party, go!’ (D2<sub>da</sub>)
- (11) a. Ako te pozovu na žurku, hoćeš li ići?  
 AKO you.DAT.SG invite.3PL.PRS on party want.2SG Q go.IMP  
 ‘If they invite you to the party, will you go?’ (D1)
- b. Ako bi te pozvali na žurku, bi li išla?  
 AKO be.3PL.AOR you.DAT.SG invited.M.PL on party be.2SG.AOR Q  
 gone.F.SG  
 ‘If they invited you to the party, would you go?’ (D2<sub>ako</sub>)
- c. Da te pozovu na žurku, bi li išla?  
 DA you.DAT.SG invite.3PL.PRS on party be.2SG.AOR Q gone.F.SG  
 ‘If they invited you to the party, would you go?’ (D2<sub>da</sub>)
- d. Da su te pozvali na žurku, bi li išla?  
 DA be.3PL.PRS you.DAT invited.M.PL on party be.2SG.AOR Q gone.F.SG  
 ‘If they had invited you to the party, would you have gone?’ (D3)

All these (and further) cases merit closer investigation in future work, but we are optimistic that many of them can be understood straightforwardly in terms of the account developed here in combination with independently motivated assumptions (e.g., see Kaufmann & Kaufmann 2021 for imperative consequents, Isaacs & Rawlins 2008 for interrogative consequents, Grosz 2012 for verbal inversion antecedents).

### 3 Analyzing Serbian X-marking

#### 3.1 The main ideas

Following standard assumptions in the linguistic literature, we assume that all conditionals *if p, q* express that all relevant *p* situations are *q* situations; they differ, however, in what counts as relevant.

To implement this, we will assume that conditional connectives (*ako, da, uko-liko, ...*) combine with the antecedent proposition *p* to introduce the set of (relevantly accessible) antecedent situations. Adopting ideas from Portner's (1997) and Yalcin's (2007) analyses of mood and epistemic modality, respectively, we take interpretation to be sensitive to a salient belief state. Serbian mood marking, that is, the distinction between INDICATIVE MOOD and CONDITIONAL MOOD, indicates whether a proposition is interpreted with respect to this belief state or (also) with respect to possibilities outside of it. By default, the salient belief state is the CONTEXT SET CS of the ongoing conversation (what is common ground between the participants).

Counterfactuals are evaluated (partly) outside of the context set. Indicative mood cannot be used in such clauses—unless the belief state parameter has been shifted. The functional element *da* is special in that it ‘meddles with mood marking’: it picks up a salient state (possibly different from the currently salient information state) and renders it the one relevant for indicative marking.<sup>5</sup> Perfect embedded in an antecedent marked with *da* signals modal expansion through ‘history rerun’ from a salient past point. We will now spell out these assumptions and apply them to Serbian X-marking conditionals, before considering possible extensions in Section 4.

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<sup>5</sup>Arguably, as a conditional connective, *da* appears in complementizer position, but the marker can spell out lower functional heads as well (Todorović & Wurmbrand 2020, a.o.). These details are irrelevant to our concerns here. We tentatively suggest that the semantic assumption carries over to *da* as heading complement clauses, though; see Section 4.

### 3.2 Serbian conditionals interpreted in a $W \times T$ -model

We assume that interpretation proceeds with respect to a model containing a set of worlds  $W$  and a set of linearly ordered temporal instants  $T$ . Variables  $w, w', w'', \dots$  and  $t, t', t'', \dots$  range over  $W$  and  $T$ , respectively.<sup>6</sup> As points of evaluation we use indices  $I = W \times T$  (variables  $i, j, \dots$ ), that is, our points of evaluation are world-time pairs (e.g.  $i = \langle w, t \rangle$ ). Declarative clauses express characterizing functions of sets of indices (simplifyingly, ‘denote sets of indices’). They are true or false at an index  $i$  and an information state  $S$ . Indices with the same world component are linearly ordered by temporal precedence, noted as  $<$  (S. Kaufmann 2005b; e.g.,  $\langle w, t \rangle < \langle w, t' \rangle$  iff  $t$  is earlier than  $t'$ ).

Interpretation depends on an index of evaluation  $i$  and a salient information state  $S$  (Portner 1997; Yalcin 2007), where  $S$  is a set of indices (noted:  $[\![\cdot]\!]^{i,S}$ ). By default, the relevant information state  $S$  is the context set  $CS$  of the ongoing conversation (that is, the set of indices compatible with what is mutual joint belief between the participants of the conversation, Stalnaker 1978; 2002).

To implement the idea that conditionals express universal quantification over suitable sets of indices, we make the following assumptions. Conditional antecedents refer to the set of indices  $j$  that (i) are relevantly accessible (indicated by  $R$ ) from the index of evaluation  $i$ , and (ii) verify the antecedent proposition  $\phi$ .<sup>7</sup>

$$(12) \quad [\![\text{If } \phi]\!]^{i,S} = \{j \in W \times T \mid [\![\phi]\!]^{j,S} \wedge j \in R(i)\}$$

The value of the accessibility relation  $R$  is context dependent and constrained by mood marking and connectives.<sup>8</sup> Once the value of  $R$  is set, at each index of evaluation  $i$  it characterizes a set of accessible indices  $j$ :

<sup>6</sup>For simplicity, we work with temporal instants; intervals could be derived as (gapless) subsets of  $T$  if needed.

<sup>7</sup>The denotation in (12) can be obtained from either of two popular accounts of conditional antecedents: (i) via abstraction over the index argument ( $\approx$  relative clause formation, e.g., Bhatt & Pancheva 2017; Arsenijević 2021) or (ii) as a plural definite of indices (e.g., Schlenker 2002; Yang 2023). Like these works, we rely on a REFERENTIAL ACCOUNT of conditionals, an alternative to QUANTIFICATIONAL ACCOUNTS on which the antecedent directly interacts with a modal operator to restrict its domain (Lewis 1975; Kratzer 1981; 2012, a.o.). We take the choice to be orthogonal to the interest of this paper. In that sense, it is not crucial that the conditional connective mediates the dependence on a contextually salient accessibility relation, it could also be treated as semantically vacuous in the sense of Kratzer (1986). It is crucial, however, that  $da$  is associated with the function of resetting the belief state parameter to the contextually salient accessibility relation, see (21).

<sup>8</sup>We abstract away from the details, but you can think of  $R$  as a free variable in the object language interpreted by a contextually given assignment.



- (13)  $R(i) = \{j \in W \times T \mid j \text{ is compatible with what } R \text{ records at index } i\}$

The consequent is predicated of the antecedent referent pointwise, yielding universal quantification (Schlenker 2002):

- (14)  $\llbracket \text{If}^R \phi, \psi \rrbracket^{i,S} = \forall i' \in \llbracket \text{If}^R \phi \rrbracket^{i,S}: \llbracket \psi \rrbracket^{i',S}$

The type of conditional expressed depends on the accessibility relation chosen. NON-PREDICTIVE INDICATIVE CONDITIONALS evaluate the consequent on the indices compatible with the speaker's belief, a subset of the indices in the current information state of the conversation (the context set, Stalnaker 1978), so  $R$  determines the set of indices that verify everything the speaker believes at a given index  $i$  ( $R_{\text{dox}}$  for 'the speaker's beliefs').

- (15) a. If Mary is working from home right now, her cat is happy.  
 b.  $\llbracket \text{If}^R \text{ Mary is working from home right now} \rrbracket^{i,S} = \{j \mid \text{WORKHOME}(\text{MARY})(j) \wedge j \in R_{\text{dox}}(i)\}$

For unembedded occurrences (matrix clause),  $R_{\text{dox}}(i)$  is a subset of the context set CS, which by default serves as the information state to which the interpretation is parameterized (so,  $R_{\text{dox}}(i) \subseteq S$ ). PREDICTIVE INDICATIVE CONDITIONALS involve antecedents that are evaluated at indices in the future of what the speakers thinks is the case currently. In this case, the accessibility relation  $R$  has to reach (a subset of) the futures of the indices reached by  $R$ , which we model by the forward expansion  $R^*$  of  $R$  (S. Kaufmann 2005b).

- (16)  $R^*(i) = \{j \mid \exists j' [j' \leq j \wedge j' \in R(i)]\}$

We assume that the antecedent of a predictive indicative conditional is interpreted with respect to  $R_{\text{dox}}^*(i)$  (the possible futures according to the speaker).

- (17) a. If Mary comes home before 6 tonight, her cat will be happy.  
 b.  $\llbracket \text{If}^R \text{ Mary comes home before 6 tonight} \rrbracket^{i,S} = \{j \mid j \in R_{\text{dox}}^*(i) \wedge \text{COMEHOMEB6}(\text{MARY})(j)\}$

X-marked conditionals (including counterfactuals and subjunctive conditionals) quantify over a set of indices obtained by CAUSAL INTERVENTION (S. Kaufmann 2013 in Kratzer-style modality; with probabilities, a.o. Kaufmann 2001; 2005a) to make room for remote or impossible antecedents. The set given by the antecedent is a superset of, or fully disjoint from, CS (the default value for the belief state parameter  $S$ ).

$$(18) \quad R^{caus}(i)(p) := \{j \mid j \text{ is reached from } i \text{ by causal intervention on variable } p\}$$

Roughly, causal intervention on the antecedent proposition  $p$  at  $i$  yields a set  $R^{caus}(i)(p)$  of indices that are co-temporal with  $i$  and retain all facts and general laws that are causally independent of  $p$ ; the world components of these indices can differ from the evaluation index  $i$  on issues that are causally determined by  $p$ ;  $i$  is the INTERVENTION TIME. (Below, we will argue that in D3-conditionals like (19), the intervention time is shifted into the past.)

- (19) a. If Mary had come home before 6 tonight, her cat would have been happy.  
 b.  $\llbracket \text{If}^R \text{ Mary had come home before 6 tonight} \rrbracket^{i,S} = \{j \mid j \in R^{caus*}(i)(\text{COMEHOME-B6(MARY)}) \wedge \text{COMEHOME-B6(MARY)}(j)\}$

Mood marking checks the status of the proposition with respect to the currently salient information state  $S$ . We take INDICATIVE MOOD to be semantically marked (Leahy 2011 for English): a sentence carrying indicative morphology is defined only on the salient information state (and its future continuations compatible with shared beliefs about the future).<sup>9</sup>

$$(20) \quad \llbracket \phi^{\text{indicative}} \rrbracket^{i,S} = \llbracket \phi \rrbracket^{i,S} \text{ if there is } i' \in S \text{ and } i' \leq i, \text{ undefined otherwise.}$$

CONDITIONAL MOOD (Serbian *bi* + PARTICIPLE) is the semantically unmarked form. Its use signals that  $R$  is interpreted to pick a domain that is not a subset of the current information state and its future continuations. This effect is an ANTIPRESUPPOSITION (Heim 1991), which in conditionals leads to the choice of  $R^{caus}$  for  $R$ .

A special twist results from the use of *DA*: unlike the other connectives (e.g. *ako*, *ukoliko*, *kada*), *da* overwrites the information state parameter for its scope with the set of indices accessible through the relation it combines with:

$$(21) \quad \llbracket da^R \phi \rrbracket^{i,S} = \{j \mid j \in R^*(i) \wedge \llbracket \phi \rrbracket^{j, \{j \mid j \in R^*(i)\}}\}$$

Thereby, the indices accessible through  $R^*$  (the forward expansion of  $R$ ) count as relevant for indicative on the proposition under *da*, explaining the occurrence of indicative in *da*-antecedents (in Sect. 4 we will suggest that these assumptions extend to *da* as introducing complement clauses).<sup>10</sup>

<sup>9</sup>For ease of exposition, our implementation omits the restriction to plausible future courses of events.

<sup>10</sup>Note that forward expansion is an identity relation on relations that already are forward expanded (i.e.,  $(R^*)^* = R^*$ ). We leave it to further research to determine if including it here is redundant for all relevant purposes.

### 3.3 Deriving Serbian X-marking

We are now in a position to explore the interaction between choice of connective and mood marking. By assumption, the domains provided by the antecedents of D2 and D3 conditionals lie (partly) outside of the overall salient information state  $S$ . In these cases, *bi*-marking on the consequent reflects that the evaluation of this proposition is not restricted to the overall salient information state  $S$  and its future (as would be presupposed by the indicative mood).

However, *ako* and *da* differ in what counts as the salient information state for the interpretation of the antecedent proposition. *ako* leaves the belief state parameter unaffected (as do other conditional connectives mentioned in Sect. 2.3); in contrast, *da* overwrites the currently salient information state with the set of indices reached through the (forward expansion of the) salient accessibility relation ((22b) repeats (21)):

- (22) a.  $\llbracket \text{ako}^R \phi \rrbracket^{i,S} = \{j \mid j \in R^*(i) \wedge \llbracket \phi \rrbracket^{j,S}\}$   
 b.  $\llbracket \text{da}^R \phi \rrbracket^{i,S} = \{j \mid j \in R^*(i) \wedge \llbracket \phi \rrbracket^{j, \{j \mid j \in R^*(i)\}}\}$

Consequently, to describe a domain (partly) outside of the current information state  $S$  (through  $R^{caus}$ ), *da* combines with a proposition marked by indicative, while *ako* requires conditional mood marking.

What remains to be explained is now the connection between D3 marking and ‘fake’ past/perfect under *da*: D3 conditionals are marked by including perfect in a *da*-marked antecedent. Serbian perfect plays the role of an all-purpose past (Browne & Alt 2004), but Todorović (2016) argues that it semantically encodes anteriority relative to a reference time (i.e., doesn’t express ‘absolute’ past w.r.t. the utterance time). Either way, that is, as past or as anteriority marker, the perfect in the antecedent of D3 conditionals seems ‘fake’ in D3 (similarly to English ‘fake’ past). The perfect in the D3 antecedent does not have to locate the event described by the antecedent before the utterance time: (23) shows that (focused) future adverbials can appear (up to speaker variation; ‘Serbian fake past’, Durović 2019):

- (23) %Da sam SUTRA prodala kola, bolje bih prošla.  
 DA be.1SG.PRS tomorrow sold.F.SG car better be.1SG.AOR passed.F.SG  
 ‘If I had sold my car TOMORROW, I would’ve gotten a better price.’

Perfect in D3 antecedents need not contribute anteriority w.r.t. the consequent either. In (23), getting the better price is part of the (hypothetical) event of selling the car, it does not follow this selling eventuality. The lack of anteriority in

Serbian D3-conditionals is evidenced also by so-called BACKTRACKERS (that is, conditionals where the consequent eventuality precedes the antecedent one and typically provides evidence for or against the antecedent). In (24), the modal state of affairs expressed in the consequent (the teleological necessity to leave before dinner) precedes Jovan's hypothetical appearance at the meeting (the antecedent eventuality, described with the perfect).

- (24) Context: You worry that John might have been at the meeting yesterday at 7 p.m. and that he might have heard the news about impending budget cuts (we hope he won't hear about it before some other decisions have been made). I aim to reassure you:

a. Sigurna sam            da Jovan sinoć        nije                bio            na  
     sure      be.3SG.PRES DA Jovan last.night not-be.3SG.PRES been.M.SG on  
     sastanku. [Da je                bio            na sastanku, morao  
     meeting. [DA be.3SG.PRES been.M.SG on meeting must.M.SG  
     bi            da ode                pre      večere], a      večerao  
     been.M.SG DA go.3SG.PRES before dinner] and had.dinner.M.SG  
     je            sa      mnom u 8. Tako da, nemoguće da je  
     be.3SG.PRES with I.INSTR in 8. So      DA impossible DA be.3SG.PRES  
     bio            tamo.  
     been.M.SG there  
     'I am sure that John wasn't at last night's meeting in the end. [If he  
     had been at the meeting, he'd have had to leave before dinner], and he  
     had dinner with me at 8 p.m. So no, he really can't have been there.'

In contrast, perfect in the antecedent of O-marked conditionals (D1) contributes either past with respect to the speech time or anteriority. The latter is evidenced in (25), in which dish washing precedes cooking at some point in the future. This is different from D1 antecedents that carry present tense, as in (26), which do not enforce anteriority; two eventualities in (26) are unordered relative to each other.

- (25) Context: You, me and Mira are flatmates and we each have chores around the house. You've been ignoring your tasks lately, so Mira and I decided that we won't do anything unless you chime in. I'm telling you what the plan is for tomorrow night.

a. Ako si                ti      do      tada oprala                sude, Mira  
     AKO be.2SG.PRES you until then wash.PFV.PART.F.SG dishes Mira  
     će      kuvati.  
     will.3 cook.INF.IMPV  
     'If you have done the dishes by then, Mira will cook.'

- (26) Context: You, me and Mira are flatmates and we each have chores around the house. I'm telling you who's doing what tomorrow night.

- a. Ako ti opereš suđe, Mira će kuvati.  
 AKO you wash.PFV.PRES.3SG dishes Mira will.3 cook.INF.IMPV  
 'If you do the dishes, Mira will cook.'

We conclude that the perfect in the *da*-antecedent is an instance of X-marking, 'fake past', in that it does not contribute temporal/aspectual meaning within the antecedent proposition. The (hypothetical) antecedent eventuality does not have to precede the speech time (i.e. the time of the evaluation of an unembedded conditional), as evidenced by *sutra* 'tomorrow' in (23); nor does it have to precede the consequent eventuality, as evidenced by (24). This is unlike what we observe for indicative conditionals (D1) as in (25).

The literature on 'fake past' ('X-marking past') is divided on whether 'fake' past is temporal ('PAST-AS-PAST' approaches, e.g. Ippolito 2013; Mirrazi & Ippolito t.a.) or modal ('PAST-AS-MODAL', e.g. Iatridou 2000; Mackay 2019). Recent works provide evidence against a uniform treatment of 'fake past' (Schulz 2017; Mizuno & Kaufmann 2019; Mizuno 2023; Kaufmann 2023): 'fake past' in English Simple Past conditionals signals modal expansion; in contrast, the second layer in English Past Perfect signals temporal backshift (semantically, PP-conditionals thus combine a genuinely temporal component with the modal shift of looking at least partly outside of the context set). First of all, the relevant works provide semantic evidence for the presence of genuinely temporal meaning in some, but not all, X-markers. By assumption, backshift past encodes causal intervention ('rerunning of history') at a salient past time *before the antecedent became settled false in the actual world*:

- (27) If I had sold my car tomorrow [rather than yesterday when I actually sold it], I would have made more money.  
 ⇒ *backshift to a moment before the salient past selling event within yesterday*

Antecedents that are not associated with a salient past point cannot be expressed with PP-X-marking, e.g. COUNTERMATHEMATICALS (see also Lewis 1973).<sup>11</sup>

- (28) a. If 9 were prime, it wouldn't be divisible by three.  
 b. #If 9 had been prime, it wouldn't have been divisible by three.

<sup>11</sup>(28b) can be acceptable if there is a salient past point at which the antecedent was under consideration, e.g., when discussing what you got wrong during a math exam you failed yesterday.

Moreover, Japanese is argued to provide morphosyntactic evidence for a split between X-marking that involves a backshift ('past-as-past') and X-marking that is purely modal (Mizuno & Kaufmann 2019). In Japanese, remoteness can, but need not be marked with ('fake') past tense (see (29) vs. (30)). However, upon closer inspection, past-marking yields the semantic equivalent of English PP-counterfactuals (D3): it is infelicitous for examples that are not naturally associated with backshifts (like the countermathematical in (31)).

- (29) Mosi [Jo-ga asita ku]-reba, [paatii-wa totemo moriagar]-u  
 ADV Jo-NOM tomorrow come-COND party-TOP very be.fun-NPST  
 daroo.  
 MODAL  
 a. 'If Jo comes tomorrow, the party will be very fun.' (D1)  
 b. 'If Jo came tomorrow, the party would be very fun.' (D2)
- (30) Mosi [Jo-ga asita ku]-reba, [paatii-wa totemo moriagat]-ta  
 ADV Jo-NOM tomorrow come-COND party-TOP very  
 daroo.  
 be.fun-PAST MODAL  
 'If Jo had come tomorrow, the party would have been very fun' (D3)
- (31) a. Mosi [kyuu-ga sosuu dear]-eba, [san-de warikir-e-na]-i.  
 ADV nine-NOM prime be-COND three-by divide-able-NEG-NPST  
 'If nine were prime, it would not be divisible by three.'  
 b. #Mosi [kyuu-ga sosuu dear]-eba, [san-de warikir-e-na]-katta.  
 ADV nine-NOM prime be-COND three-by divide-able-NEG-PAST  
 #'If nine had been prime, it would not have been divisible by three.'

Semantically, Serbian D3-conditionals pattern with English PP-counterfactuals and Japanese past-marked counterfactuals. Atemporal countermathematicals are expressed with D2-conditionals ((32) with *da* or *ako*), D3 is infelicitous (33):<sup>12</sup>

- (32) a. Da je 9 prost broj, ne bi bio deljiv  
 da be.3SG.PRS 9 simple number not be.3SG.AOR been.M.SG divisible  
 sa 3.  
 with 3

<sup>12</sup>The Serbian speakers we consulted resist the strategies that render countermathematicals in D3-counterfactuals marginally acceptable in English or Japanese (see (28b) above and Mizuno & Kaufmann 2019). One speaker reports that (33) 'sounds like 9 died', which means that the antecedent proposition is no longer interpreted as an atemporal mathematical fact. We thank Una Stojnić (p.c.) for pointing out an error in our previous discussion of this point.

- b. Ako **bi** 9 **bio** prost broj, ne **bi** **bio**  
 ako be.3SG.AOR 9 been.M.SG simple number not be.3SG.AOR been.M.SG  
 deljiv sa 3.  
 divisible with 3  
 ‘If 9 were prime, it would not be divisible by 3.’ (D2)

- (33) #Da **je** 9 **bio** prost broj, ne **bi** **bio**  
 da be.3SG.PRS 9 been.M.SG simple number not be.3SG.AOR been.M.SG  
 deljiv sa 3.  
 divisible with 3  
 ‘If 9 had been prime, it would have not been divisible by 3.’ (D3)

To account for the semantic effect, we take D3 to be interpreted like D2, but embedded under temporal backshift to ‘rerun history’ from a past intervention point:

- (34) a. If Mary had come home before 6, her cat would have been happy.  
 b.  $\exists i'[i' < i \wedge$   
 $\forall j[[j \in R^{caus*}(i')(\text{COMEHOME-B6}(\text{MARY})) \wedge$   
 $\text{COMEHOME-B6}(\text{MARY})(j)] \rightarrow \text{HAPPY}(\text{CAT-OF-MARY})(j')]]$

Note that this leaves us with a COMPOSITIONALITY PUZZLE: PERFECT within antecedent seems to scope over the counterfactual semantically. Similar problems result for English, see Ippolito (2013) (in contrast to the scopally transparent Japanese).<sup>13</sup>

Further evidence for semantically encoded backshift in D3 (past-as-past) comes from the interaction with temporal adverbials (after Japanese examples in Mizuno 2023). Antecedents of D2-conditionals cannot talk about indices in the past (neither with *ako*, (35), nor with *da*, (36)).

- (35) \*Ako **bi** Miloje otišao na žurku juče, žurka **bi**  
 AKO be.3SG.AOR Miloje gone.M.SG on party yesterday party be.3SG.AOR  
 uspela.  
 succeeded.F.SG

<sup>13</sup> Arregui (2005; 2007) analyzes the additional perfect in PP conditionals as an aspect marker that yields a crucial difference to perfective aspect in SP conditionals, but Ippolito (2013) points out significant problems with her account. The independence between aspectual oppositions and X-marking in Serbian (cf. (6) and (7)) provides further evidence against Arregui’s account. Groenn 2021 provides an explicit morphosyntactic analysis for English SP and PP conditionals, where the additional perfect marking in PP antecedents is licensed under agreement with a backshift operator outside of the conditional, resulting in a structure similar to what we are proposing for Serbian.

- (36) \*Da Miloje ode na žurku juče, žurka bi  
DA Miloje go.3SG.PRS on party yesterday party be.3SG.AOR  
uspela.  
succeeded.F.SG  
Int.: ‘If Miloje were to have been on the party yesterday, the party would have been successful.’

In contrast, D3-conditionals can describe past eventualities (as well as present or future ones, see e.g. (23)):

- (37) Da je Miloje otišao na žurku juče, žurka bi  
DA be.3SG.PRS Miloje gone.M.SG on party yesterday party be.3SG.AOR  
uspela.  
succeeded.F.SG  
‘If Miloje had been at the party yesterday, the party would have been successful.’

This suggests that the backshift encoded by the perfect in the *da*-antecedent does indeed take evaluation back to a past branching point and allows for quantification over the alternative histories that extend from there.

## 4 Possible implications beyond conditional clauses

### 4.1 *da*-complements and *da*-purposive clauses

Connective *da* itself triggers (i) forward expansion (like any conditional connective), and (ii) overwrites what counts as the salient belief state (the domain of indicative mood). In this respect, it is relevant to point out that Serbian *da*-complements with indicative morphology appear also under predicate types that cross-linguistically tend to take infinitival (Serbian: optional) or subjunctive complements (e.g. WANT- or TRY-predicates) (in addition to appearing in non-factive indicative complements like under SAY or BELIEVE; Browne & Alt 2004; Todorović & Wurmbrand 2020; Kaufmann et al. 2023).

- (38) a. Petra želi trčati.  
Petra want.3SG.PRS run.INF  
‘Petra wants to run.’  
b. Petra želi da trči.  
Petra want.3SG.PRS DA run.3SG.PRS  
‘Petra wants to run.’



We assume that *da*'s *R* variable is anaphoric to the modality of the matrix clause in such cases.

*da*-clauses also realize PURPOSIVE CLAUSES, where they can optionally be indicative or conditional marked ([Arsenijević 2021](#), his (24)):

- (39) Polomio je staklo da { uskoči / bi  
 broken.M.SG be.3SG.PRS glass DA jump-in.3SG.PRS / be.1SG.AOR  
 uskočio } u sobu.  
 jumped-in.M.SG in room  
 'He broke the glass in order to jump into the room.'

We consider this optionality evidence that purposive clauses are sensitive to two different information states: (i) the belief state of the agent, and (ii) the state where all the agent's goals are realized (the GOAL STATE). The sentence expresses that the goal proposition ('that he jumps into the room') is true throughout the goal state, which need not be a subset of the overall salient information state (some of the agent's goals may be unobtainable). If *da*'s contextual variable *R* is set to this goal state, the purposive clause carries indicative; if *da*'s contextual variable *R* is set to the agent's belief state, the purposive clause carries conditional mood (*bi* + past participle). (See [Sæbø 1991](#) for purposive constructions; [Nissenbaum 2005](#) for a compositional analysis of English examples.)

## 4.2 Standalone uses for directives and optatives

*da*- and *ako*-clauses can also appear as standalone matrix clauses. Both *da*-clauses (discussed in [Grosz 2012](#)) and *ako*-clauses can serve as OPTATIVES:

- (40) Da Jovan { makar / samo } poslušaj Mariju!  
 DA Jovan { at.least / only } listen-to.3SG.PRS Marija  
 'If only Jovan were to listen to Marija!'
- (41) Da je Jovan { makar / samo } poslušao Mariju!  
 DA be.3SG Jovan { at.least / only } listened-to.M.SG Marija  
 'If only John had listened to Mary!' ([Grosz 2012](#): 43c/536d)
- (42) Ako bi me samo poslušala!  
 AKO be.3SG.AOR I.ACC only listened-to.F.SG  
 'If only you were to listen to me!'

[Grosz \(2012\)](#) discusses standalone uses of *if* and *that*-clauses as optatives and exclamatives, exemplified in (43):

- (43) a. Oh, that I had told them both a year ago!  
(Martin F. Tupper. 1851. *The Twins; A Domestic Novel*, Grosz 2012: 1b)  
b. If only I had told them both a year ago! (Grosz 2012: 1b)

Grosz assumes that these standalone occurrences express propositions that are marked for mood (for him, indicating directly a relation to the context set). They combine with a COVERT EXCLAMATIVE OPERATOR that expresses that the proposition surpasses a contextually given threshold for desirability (optatives) (or unexpectedness in exclamatives; interestingly, *ako* does not express exclamatives, *da* remains to be tested more carefully.)<sup>14</sup>

In addition, for *da*-clauses, we observe DIRECTIVE READINGS (Vrzić 1996; Kaufmann et al. 2023; Oikonomou & Ilić 2023).

- (44) Da pročitáš      ovu knjigu!  
DA read.2SG.PRS this book  
'Read this book (already)!'

We assume that there is a covert prioritizing (deontic or bouletic) necessity modal (Kaufmann et al. 2023) that combines with the *da*-marked proposition and determines the value of the accessibility relation *R*.

## 5 Conclusions

We have proposed a compositional analysis of the interplay of conditional connective, mood, and perfect ('fake past') as encoding different types of remoteness in Serbian conditionals. Mood marking indicates sensitivity to the contextually salient belief state. Indicative is marked (a proposition expressed by a clause bearing indicative mood is restricted to interpretation on the contextually salient belief state); conditional mood is unmarked. Conditional connectives are sensitive to contextually salient accessibility relations, and conditional antecedents refer to the set of indices that, at the index of evaluation, are accessible by the given relation and verify the antecedent proposition. The connective *da* interacts with the information state parameter; it resets it to the modality *da* is sensitive to. The other conditional connectives (*ako*,...) do not affect the information state parameter. Perfect in the antecedent of *da* conditionals indicates backshift to a salient

<sup>14</sup>Grosz (2012: Sect. 4.1.) argues against an alternative analysis in terms of a deleted matrix clause. While we find his arguments convincing, our considerations would carry over to a deletion analysis, which assimilates the data discussed in this section to what was discussed in the previous one.

point in the past at which causal intervention is performed to make accessible alternative histories.

The Serbian data fit the semantic characterization of (co-temporal) modal expansion (D2) and backshift to ‘rerun history’ (D3) as hypothesized in the formal literature (Mizuno & Kaufmann 2019; Mizuno 2023; Kaufmann 2023). Serbian is particularly interesting as it fills a gap in the crosslinguistic picture described so far: it distinguishes *transparently* between modal X-marking and temporal X-marking. This is not the case for any of the languages that have been discussed previously with respect to this split: English uses ‘fake’ past/perfect for both types; Japanese leaves D2 unmarked and marks D3 (backshift) with past tense; Slovenian only marks modal expansion but leaves backshift unmarked (Mizuno 2023), cf. (45).

- (45) Če bi Rok imel denar, bi ti ga posodil.  
 if BI Rok have.PART.3.SG.M money BI to.you it lend.PART.3.SG.M  
 ‘If Rok had any money, he would lend it to you.’ (Mizuno 2023: 3a)

The Serbian pattern provides strong empirical support for a distinction that has been drawn based on semantic contrasts and overt instantiations of parts of the pattern in other languages.

We have also shown possible extensions of our assumptions about Serbian mood marking and the role of connective *da*, which enables indicative marked *da*-complement clauses to play the role of subjunctives in other languages. Our account naturally extends to standalone occurrences of connective marked clauses as well as the optionality of conditional mood marking in purposive clauses: if these are assumed to be sensitive to both beliefs and goals, making either aspect available to guide mood marking.

## Abbreviations

|      |              |     |                   |
|------|--------------|-----|-------------------|
| 1    | first person | INS | instrumental      |
| 3    | third person | M   | masculine         |
| AOR  | aorist       | NOM | nominative        |
| ACC  | accusative   | PFV | perfective        |
| DAT  | dative       | PL  | plural            |
| F    | feminine     | PRS | present tense     |
| IPFV | imperfective | Q   | question particle |
| INF  | infinitive   | SG  | singular          |

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